# Switch-mode Power Rectifier

## D²PAK Surface Mount Power Package

# MBRB1545CTG, SBRB1545CTG

The D²PAK Power Rectifier employs the Schottky Barrier principle with a platinum barrier metal. These state-of-the-art devices have the following features:

### Features

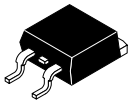
- Center−Tap Configuration
- Guardring for Stress Protection
- Low Forward Voltage
- 175°C Operating Junction Temperature
- Epoxy Meets UL 94, V−0 @ 0.125 in
- Short Heatsink Tab Manufactured − Not Sheared
- Similar in Size to the Industry Standard TO220 Package
- SBRB Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC−Q101 Qualified and PPAP Capable
- These Devices are Pb−Free and are RoHS Compliant*

### Mechanical Characteristics

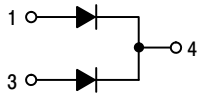
- Case: Epoxy, Molded, Epoxy Meets UL 94, V−0
- Weight: 1.7 grams (approximately)
- Finish: All External Surfaces Corrosion Resistant and Terminal Leads are Readily Solderable
- Lead and Mounting Surface Temperature for Soldering Purposes: 260°C Max. for 10 Seconds
- Device Meets MSL1 Requirements
- ESD Ratings:
  - Machine Model = C (> 400 V)
  - Human Body Model = 3B (> 8000 V)

*For additional information on our Pb-Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

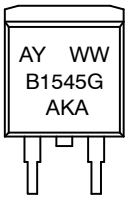
**SCHOTTKY BARRIER RECTIFIER 15 AMPERES 45 VOLTS**



**D²PAK**
**CASE 418B**
**STYLE 3**




### MARKING DIAGRAM




| | |
|---|---|
| B1545 | = Device Code |
| A | = Assembly Location |
| Y | = Year |
| WW | = Work Week |
| G | = Pb−Free Package |
| AKA | = Diode Polarity |

### ORDERING INFORMATION

See detailed ordering and shipping information on page 3 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 3.

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**MAXIMUM RATINGS** (Per Leg)

| Symbol | Rating | Value | Unit |
|---|---|---|---|
| $V_{RRM}$<br>$V_{RWM}$<br>$V_R$ | Peak Repetitive Reverse Voltage<br>Working Peak Reverse Voltage<br>DC Blocking Voltage | 45 | V |
| $I_{F(AV)}$ | Average Rectified Forward Current<br>(Rated $V_R$, $T_C$ = 167°C) Total Device | 7.5<br>15 | A |
| $I_{FRM}$ | Peak Repetitive Forward Current<br>(Rated $V_R$, Square Wave, 20 kHz, $T_C$ = 166°C) | 15 | A |
| $I_{FSM}$ | Non−Repetitive Peak Surge Current<br>(Surge Applied at Rated Load Conditions Halfwave, Single Phase, 60 Hz) | 150 | A |
| $I_{RRM}$ | Peak Repetitive Reverse Surge Current (2.0 μs, 1.0 kHz) | 1.0 | A |
| $T_{stg}$ | Storage Temperature Range | −65 to +175 | °C |
| $T_J$ | Operating Junction Temperature (Note 1) | −65 to +175 | °C |
| dv/dt | Voltage Rate of Change (Rated $V_R$) | 10,000 | V/μs |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. The heat generated must be less than the thermal conductivity from Junction−to−Ambient: $dP_D/dT_J < 1/R_{\theta JA}$.

**THERMAL CHARACTERISTICS** (Per Leg)

| Symbol | Characteristic | Value | Unit |
|---|---|---|---|
| | Thermal Resistance | | °C/W |
| $R_{\theta JC}$ | Junction to Case | 2.0 | |
| $R_{\theta JA}$ | Junction to Ambient (Note 2) | 50 | |

2. When mounted using minimum recommended pad size on FR−4 board.

**ELECTRICAL CHARACTERISTICS** (Per Leg)

| Symbol | Characteristic | Value | Unit |
|---|---|---|---|
| $V_F$ | Maximum Instantaneous Forward Voltage (Note 3) | | V |
| | ($i_F$ = 7.5 Amps, $T_J$ = 125°C) | 0.57 | |
| | ($i_F$ = 15 Amps, $T_J$ = 125°C) | 0.72 | |
| | ($i_F$ = 15 Amps, $T_J$ = 25°C) | 0.84 | |
| $i_R$ | Maximum Instantaneous Reverse Current (Note 3) | | mA |
| | (Rated dc Voltage, $T_J$ = 125°C) | 15 | |
| | (Rated dc Voltage, $T_J$ = 25°C) | 0.1 | |

3. Pulse Test: Pulse Width = 300 μs, Duty Cycle ≤ 2.0%.



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## TYPICAL CHARACTERISTICS

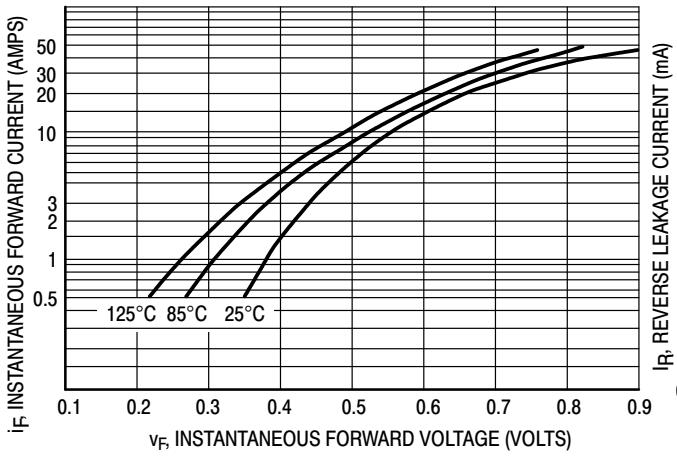



**Figure 1. Typical Forward Voltage, Per Leg**

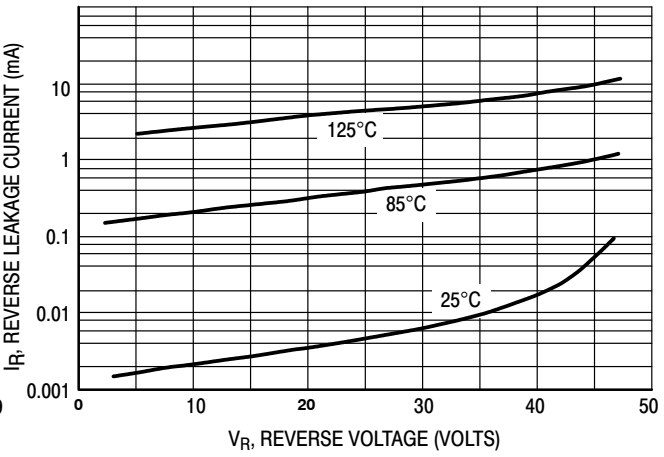



**Figure 2. Typical Reverse Current, Per Leg**

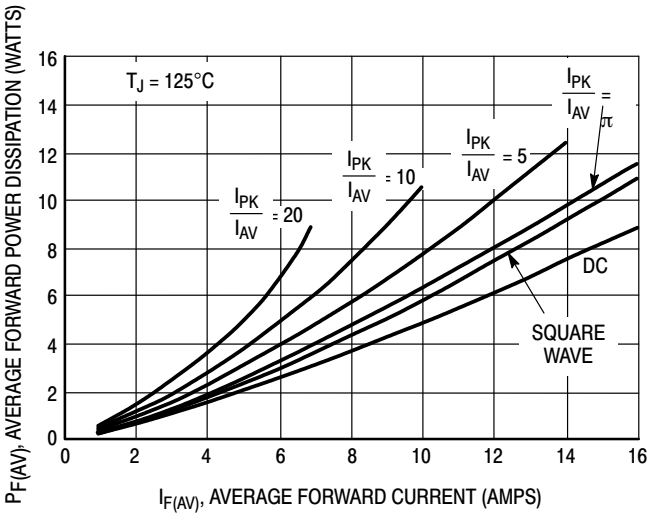



**Figure 3. Typical Forward Power Dissipation**

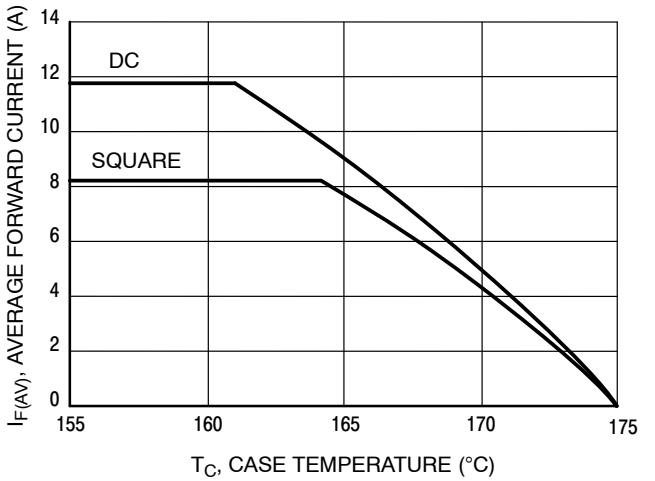



**Figure 4. Current Derating, Case, Per Leg**

### ORDERING INFORMATION

| Device | Package | Shipping† |
|---|---|---|
| MBRB1545CTT4G | D²PAK (Pb−Free) | 800 / Tape & Reel |
| SBRB1545CTT4G | D²PAK (Pb−Free) | 800 / Tape & Reel |

**DISCONTINUED** (Note 4)

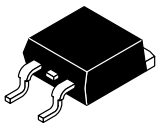
| | | |
|---|---|---|
| MBRB1545CTG | D²PAK (Pb−Free) | 50 Units / Rail |
| SBRB1545CTG | D²PAK (Pb−Free) | 50 Units / Rail |

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

4. **DISCONTINUED:** These devices are not recommended for new design. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.



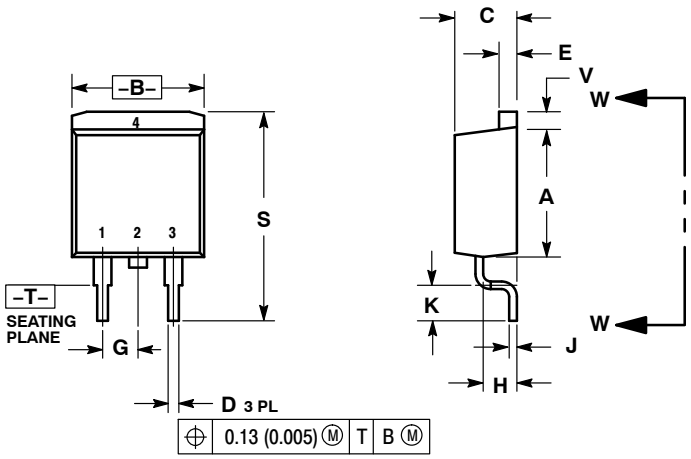
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**SCALE 1:1**

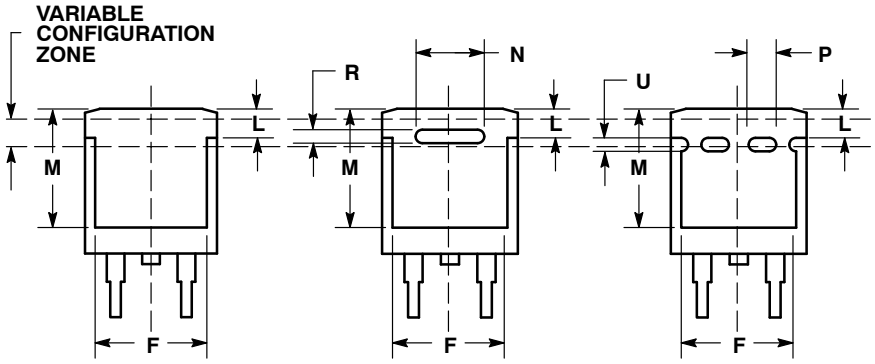
**D²PAK 3**
CASE 418B–04
ISSUE L

DATE 17 FEB 2015




NOTES:
1. DIMENSIONING AND TOLERANCING PER ANSI Y14.5M, 1982.
2. CONTROLLING DIMENSION: INCH.
3. 418B–01 THRU 418B–03 OBSOLETE, NEW STANDARD 418B–04.

| | INCHES | | MILLIMETERS | |
|---|---|---|---|---|
| DIM | MIN | MAX | MIN | MAX |
| A | 0.340 | 0.380 | 8.64 | 9.65 |
| B | 0.380 | 0.405 | 9.65 | 10.29 |
| C | 0.160 | 0.190 | 4.06 | 4.83 |
| D | 0.020 | 0.035 | 0.51 | 0.89 |
| E | 0.045 | 0.055 | 1.14 | 1.40 |
| F | 0.310 | 0.350 | 7.87 | 8.89 |
| G | 0.100 BSC | | 2.54 BSC | |
| H | 0.080 | 0.110 | 2.03 | 2.79 |
| J | 0.018 | 0.025 | 0.46 | 0.64 |
| K | 0.090 | 0.110 | 2.29 | 2.79 |
| L | 0.052 | 0.072 | 1.32 | 1.83 |
| M | 0.280 | 0.320 | 7.11 | 8.13 |
| N | 0.197 REF | | 5.00 REF | |
| P | 0.079 REF | | 2.00 REF | |
| R | 0.039 REF | | 0.99 REF | |
| S | 0.575 | 0.625 | 14.60 | 15.88 |
| V | 0.045 | 0.055 | 1.14 | 1.40 |




VIEW W–W 1 | VIEW W–W 2 | VIEW W–W 3

STYLE 1:
PIN 1. BASE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 2:
PIN 1. GATE
2. DRAIN
3. SOURCE
4. DRAIN

STYLE 3:
PIN 1. ANODE
2. CATHODE
3. ANODE
4. CATHODE

STYLE 4:
PIN 1. GATE
2. COLLECTOR
3. EMITTER
4. COLLECTOR

STYLE 5:
PIN 1. CATHODE
2. ANODE
3. CATHODE
4. ANODE

STYLE 6:
PIN 1. NO CONNECT
2. CATHODE
3. ANODE
4. CATHODE

**MARKING INFORMATION AND FOOTPRINT ON PAGE 2**



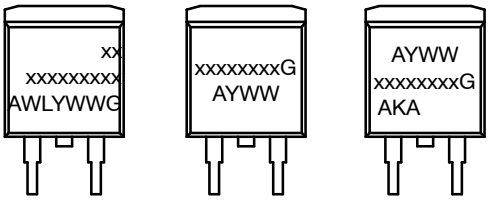
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**D²PAK 3**
CASE 418B−04
ISSUE L

DATE 17 FEB 2015

## GENERIC MARKING DIAGRAM*

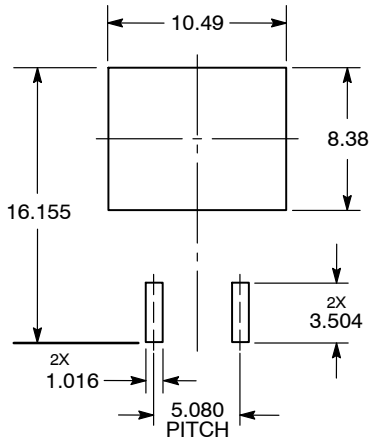



**IC** **Standard** **Rectifier**

| | |
|---|---|
| xx | = Specific Device Code |
| A | = Assembly Location |
| WL | = Wafer Lot |
| Y | = Year |
| WW | = Work Week |
| G | = Pb−Free Package |
| AKA | = Polarity Indicator |

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

## SOLDERING FOOTPRINT*




DIMENSIONS: MILLIMETERS

*For additional information on our Pb−Free strategy and soldering details, please download the **onsemi** Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.




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